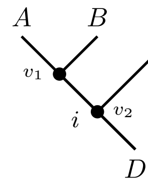
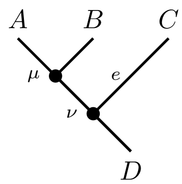
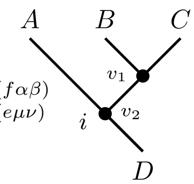
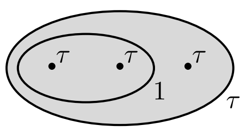
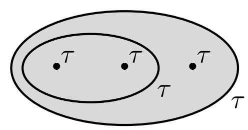
$$\mathrm{Hom}(A\otimes B\otimes C, D) = \mathrm{span}\,\{$$  $$\}$$

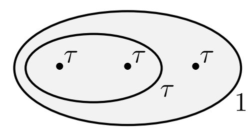


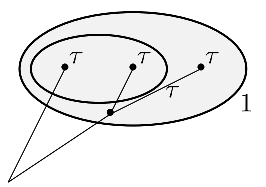
$$= \sum_{f,\alpha,\beta} [F_D^{ABC}]^{(f\alpha\beta)}_{(e\mu\nu)}$$



$|0\rangle =$ 

$|1\rangle =$ 

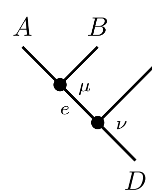
$|NC\rangle =$ 



$$\begin{tikzcd}
(A\otimes B)\otimes C \arrow[rr, "\alpha_{ABC}"] \arrow[d, "Hom(-,D)"] & \text{Yoneda-lifting} & A\otimes(B\otimes C) \arrow[d, "Hom(-,D)"] \\
Hom((A\otimes B)\otimes C, D) \arrow[d, phantom, "\cong" description, "\text{semisimplicity}"] & \cong & Hom(A\otimes(B\otimes C), D) \arrow[d, phantom, "\cong" description, "\text{semisimplicity}"] \\
\bigoplus_c Hom(A\otimes B, c)\otimes Hom(c\otimes C, D) \arrow[d, phantom, "\cong" description] & \cong & \bigoplus_{d\in Irr(\mathsf{Cat})} Hom(A\otimes d, D)\otimes Hom(B\otimes C, d \arrow[d, phantom, "\cong" description] \\
\bigoplus_c V_c^{AB}\otimes V_D^{cC} \arrow[d, phantom, "\in" description] & \text{choice of basis} & \bigoplus_c V_D^{Ac}\otimes V_c^{BC} \arrow[d, phantom, "\in" description] \\
|(A,B);c;\mu\rangle\otimes|c,C;D;\nu\rangle \arrow[rr, equal] & & \sum_{f,\alpha,\beta}[F_D^{ABC}]^{(f\alpha\beta)}_{(c\mu\nu)}|A,f;D;\beta\rangle\otimes|B,C;f;
\end{tikzcd}$$

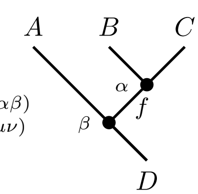
Let $f : E \to A \otimes B$ and $g : D \to E \otimes C$.

$$
\begin{tikzcd}
(A \otimes B) \otimes C \arrow[rr, "\overset{\alpha_{ABC}}{\sim}"] \arrow[d, mapsto, "Hom(-,D)"] & & A \otimes (B \otimes C) \arrow[d, mapsto, "Hom(-,D)"] \\
Hom((A \otimes B) \otimes C, D) \arrow[rr, "\overset{\alpha_{A,B,C}}{\sim}", "\text{associator induces isomorphims under yoneda}"'] \arrow[d, phantom, "\text{semisimplicity}\ \cong"] & & Hom(A \otimes (B \otimes C), D) \arrow[d, phantom, "\text{semisimplicity}\ \cong"] \\
\bigoplus_{c \in Irr(CAT)} Hom(A \otimes B, c) \otimes Hom(c \otimes C, D) \arrow[rr, phantom, "\cong"] \arrow[d, phantom, "="] & & \bigoplus_{d \in Irr(\mathsf{Cat})} Hom(A \otimes d, D) \otimes Hom(B \otimes C, d) \arrow[d, phantom, "="] \\
\bigoplus_c V_c^{AB} \otimes V_D^{cC} \arrow[rr, phantom, "\cong"] \arrow[d, phantom, "\Cap"] & & \bigoplus_c V_D^{Ac} \otimes V_c^{BC} \arrow[d, phantom, "\Cap"] \\
\underset{\mu \in 1,\dots,N_c^{AB}}{|(A,B);c;\mu\rangle} \otimes \underset{\nu \in 1,\dots,N^{cC,D}}{|c,C;D;\nu\rangle} \arrow[rr, "F_D^{ABC}", "\text{6j-symbols}"'] & & \sum_{f,\alpha,\beta} [F_D^{ABC}]_{(c\mu\nu)}^{(f\alpha\beta)} |A,f;D;\beta\rangle \otimes |B,C;f;\alpha\rangle
\end{tikzcd}
$$



$=$

$$\sum_{f,\alpha,\beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)}$$



$$
\begin{tikzcd}
(A \otimes B) \otimes C \arrow[r, "\alpha_{ABC}", "\sim"'] \arrow[d, "Hom(-,D)"] & A \otimes (B \otimes C) \arrow[d, "Hom(-,D)"] \\
Hom((A \otimes B) \otimes C, D) \arrow[r, "\alpha_{A,B,C}", "\text{associator induces isomorphims under yoneda}"'] \arrow[d, phantom, "\text{semisimplicity}\ \cong"] & Hom(A \otimes (B \otimes C), D) \arrow[d, phantom, "\text{semisimplicity}\ \cong"] \\
\bigoplus_{c \in Irr(CAT)} Hom(A \otimes B, c) \otimes Hom(c \otimes C, D) \arrow[r, phantom, "\cong"] \arrow[d, phantom, "="] & \bigoplus_{d \in Irr(\mathsf{Cat})} Hom(A \otimes d, D) \otimes Hom(B \otimes C, d) \arrow[d, phantom, "="] \\
\bigoplus_{e} V_e^{AB} \otimes V_D^{eC} \arrow[r, phantom, "\cong"] \arrow[d, phantom, "\ni"] & \bigoplus_{f} V_D^{Af} \otimes V_f^{BC} \arrow[d, phantom, "\ni"] \\
\underset{\mu \in 1,\dots,N_e^{AB}}{|(A,B); e; \mu\rangle} \otimes \underset{\nu \in 1,\dots,N^{eC,D}}{|e, C; D; \nu\rangle} \arrow[r, "F_D^{ABC}", "\text{6j-symbols}"'] & \sum_{f,\alpha,\beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle
\end{tikzcd}
$$

$$
{}^{\alpha_3}_{\substack{\alpha_2 \\ \alpha_1}}\Big[F^{D}_{A\,B\,C}\Big]^{\beta_3}_{\substack{\beta_2 \\ \beta_1}}
$$